

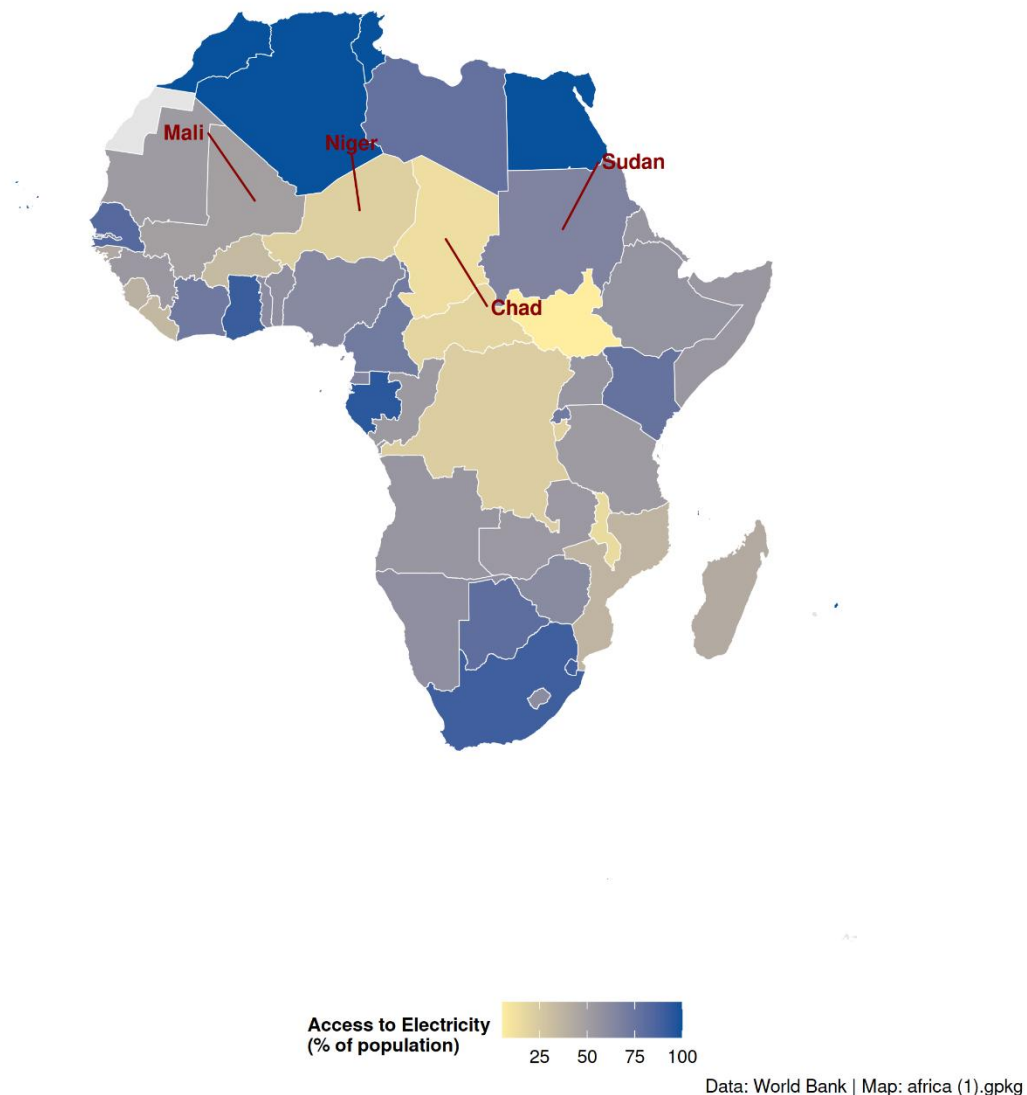
1.OBJECTIVE

Access to electricity in Africa is sharply divided by the Sahara Desert. North African countries have achieved near-universal electrification, but the Sahel region directly to the south experiences a massive, sudden drop in access. This highlights how extreme physical geography creates a hard boundary for basic infrastructure development.

2.GRAPHIC

The Geographic Divide of Energy Poverty in Africa (2024)

Highlighting the Sahel region as the infrastructural boundary of electrification



3.NARRATIVE

This choropleth map visualizes electricity access across Africa using 2024 World Bank data. Applying Tufte's design principles, I removed non-essential elements like latitudinal gridlines and background panels to maximize the data-ink ratio and let the regional contrast stand out. (Gray regions represent areas with missing data).

The map shows a clear geographic divide rather than a random distribution of energy poverty. To illustrate this insight, I annotated four Sahel states: Mali, Niger, Chad and Sudan. These

countries sit right at the edge of the Sahara, where the harsh terrain makes expanding national power grids incredibly difficult and expensive. The sharp transition from 100% access in the north to rates as low as 13% in Chad proves that physical geography acts as a primary bottleneck for African infrastructure and broader economic growth.

4.CODE

```
# 1. Load packages
> library(sf)
> library(dplyr)
> library(ggplot2)
> library(ggrepel)

> > # 2. Import the base map
> africa_map <- st_read("africa (1).gpkg", layer = "all territories")

> > # 3. Read the electricity data (Index 7)
> elec_data <- read.csv("electricity.csv", na.strings = c("..", "")) %>%
+   select(Country.Code = 2, Electricity_Access = 7) %>%
+   filter(!is.na(Electricity_Access))

> > # 4. Merge the map and data
> map_data <- africa_map %>%
+   left_join(elec_data, by = c("ISO" = "Country.Code"))

> > # 5. Isolate the countries
> sahel_epicenters <- map_data %>%
+   filter(NAME_0 %in% c("Mali", "Niger", "Chad", "Sudan"))

> > # 6. Generate the map
> pub_map <- ggplot(map_data) +
+   # Base choropleth with thin white borders to reduce non-data ink
+   geom_sf(aes(fill = Electricity_Access), color = "white", linewidth = 0.2) +
+
+   # ggrepel pushes labels into empty space and draws a line to the country
+   geom_text_repel(
+     data = sahel_epicenters,
+     aes(label = NAME_0, geometry = geom),
+     stat = "sf_coordinates",
+     color = "darkred",
+     fontface = "bold",
+     size = 3.5,
+     box.padding = 2,           # Pushes the text into the white space
+     min.segment.length = 0,    # Forces the connecting line to draw
+     segment.color = "darkred",
+     segment.linewidth = 0.6
+   ) +
+
+   # Academic styling (yellow-to-dark-blue palette is great for electricity)
+   scale_fill_gradient(low = "#ffeda0", high = "#08519c", na.value = "gray90"),
+   name = "Access to Electricity\n(% of population)" +
+   labs(
```

```

+       title = "The Geographic Divide of Energy Poverty in Africa (2024)",
+       subtitle = "Highlighting the Sahel region as the infrastructural bou
ndary of electrification",
+       caption = "Data: World Bank | Map: africa (1).gpkg"
+     ) +
+
+     # Strict Tufte compliance: theme_void() removes all coordinates, axes, a
nd background panels
+     theme_void() +
+     theme(
+       plot.title = element_text(face = "bold", size = 13, hjust = 0.5),
+       plot.subtitle = element_text(size = 10, face = "italic", hjust = 0.5
),
+       legend.position = "bottom",
+       legend.title = element_text(size = 9, face = "bold")
+     )

```

Warning message:

In geom_text_repel(data = sahel_epicenters, aes(label = NAME_0, :

Ignoring unknown parameters: `segment.linewidth`

```
> > # 7. Save the map
```

```
> ggsave("africa_electricity_map_2024.png", plot = pub_map, width = 8, height
= 8, dpi = 300, bg = "white")
```

NB: Red text is generated by AI(Gemini)